

Integration of the FINS Protocol into the Open-Source HMI System FUXA

Denny Chrisnanda Frans Sebastian Gilang Hansagita

Bina Nusantara University

IEEE Conference Presentation

- Motivation and Problem Statement
- Research Questions and Contributions
- Related Works
- System Architecture and Methodology
- Implementation and Experimental Setup
- Results and Discussion
- Conclusions and Future Work

- Omron PLCs dominate production lines at PT XYZ ($> 90\%$ machines).
- Factory Interface Network Service (FINS) is the de facto Omron protocol.
- Commercial HMI/SCADA systems are costly and vendor-locked.
- Open-source HMI FUXA lacks native FINS support.

Problem Statement

- Lack of native FINS support prevents FUXA adoption in Omron-centric plants.
- Middleware-based solutions increase latency and complexity.
- Alarm and real-time visualization reliability are critical requirements.

RQ1

How can FINS protocol support be implemented in FUXA with reliability and feature parity comparable to built-in protocols?

RQ2

How can stable, low-latency read/write operations be achieved for real-time visualization?

Main Contributions

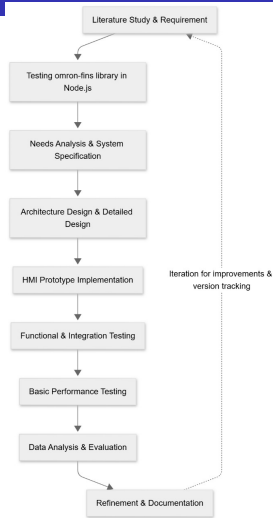
- First native integration of Omron FINS into the FUXA HMI platform.
- Backend driver (`DeviceFins`) and frontend configuration support.
- Empirical evaluation of latency, reliability, resource usage, and alarms.
- Identification of alarm latency bottleneck in polling-based HMI systems.

Related Works I

Author & Year	Platform	Domain	Protocol	Contribution
Uddin et al. (2022)	Node-RED, Grafana	Solar RO plant	HTTP, MQTT	Open-source SCADA feasibility
Omidi et al. (2023)	Node-RED SCADA	Renewable energy	MQTT, REST	Modular real-time SCADA
Almas et al. (2014)	Open SCADA	Smart grid	DNP3	Polling performance analysis
Hazdi et al. (2017)	Web SCADA	Home automation	OPC	Web-based PLC SCADA
Arnoud et al. (2021)	FUXA	Cybersecurity	Modbus TCP	FUXA as HMI testbed
Salazar et al. (2024)	ICSNet + FUXA	Industrial honeynet	Multi-protocol	ICS security visualization
Gil et al. (2022)	Framework	Industrial systems	Multi-protocol	Interoperability analysis
Giehl et al. (2024)	OpenPLC	ICS	OPC UA	Secure OPC UA integration
This work	FUXA	Manufacturing	Omron FINS	First native FINS integration

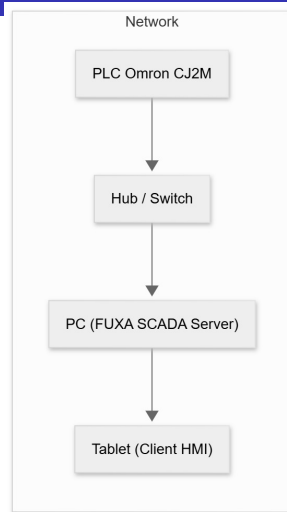
- Existing studies focus on Modbus, OPC UA, MQTT.
- No native open-source HMI support for Omron FINS.
- Alarm latency rarely analyzed explicitly.

- Iterative development:
 - Analysis
 - Design
 - Implementation
 - Evaluation



- Omron PLC
- Node.js backend (FUXA + DeviceFins)
- Angular frontend (HMI)

Architecture Diagram



Implementation

- Node.js DeviceFins driver
- FINS UDP/TCP support
- PLC memory access (DM, CIO, W)
- Auto reconnect and timeout handling

- Device configuration UI
- Tag editor
- Event-based real-time updates

Performance Results

- Latency: **12 ms**
- Success rate: **99%**
- CPU usage: **2%**
- Memory usage: **462 MB**

Metric	KPI Target
Latency	≤ 200 ms
Success Rate	$\geq 95\%$
CPU Usage	$\leq 30\%$
Memory Usage	≤ 500 MB
Alarm Detection	≤ 200 ms
Startup Time	≤ 10 s

Metric	KPI Target
Latency	≤ 200 ms
Success Rate	$\geq 95\%$
CPU Usage	$\leq 30\%$
Memory Usage	≤ 500 MB
Alarm Detection	≤ 200 ms
Startup Time	≤ 10 s

- Detection delay: **1000 ms**
- Cause: 1-second polling interval

- First native FINS integration in FUXA
- Reliable low-latency communication
- Alarm latency is main limitation

Thank You

Questions?